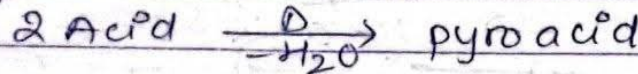
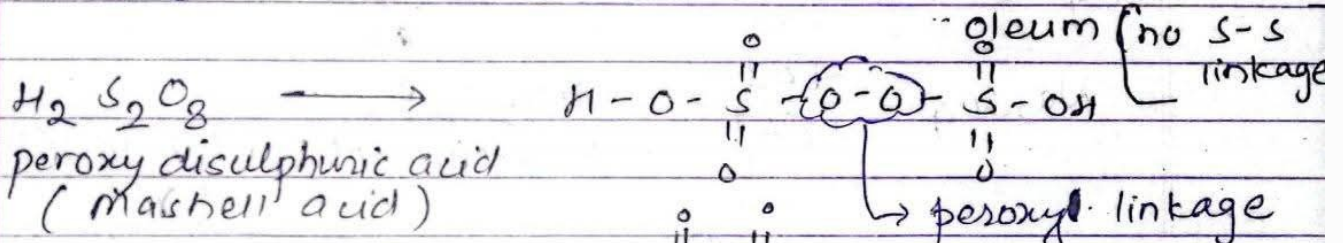
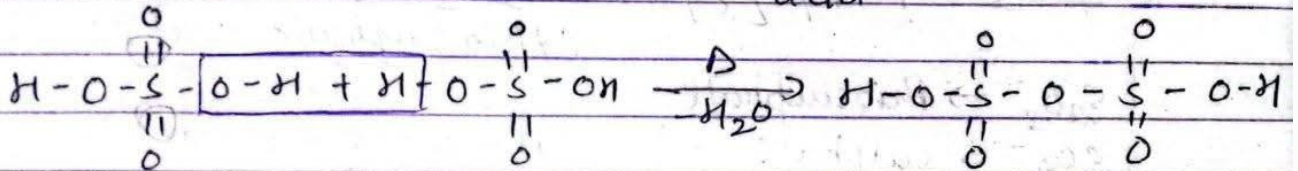
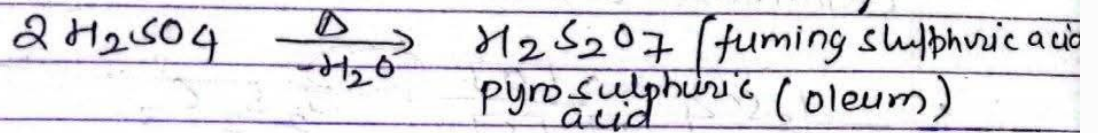


Inorganic basic

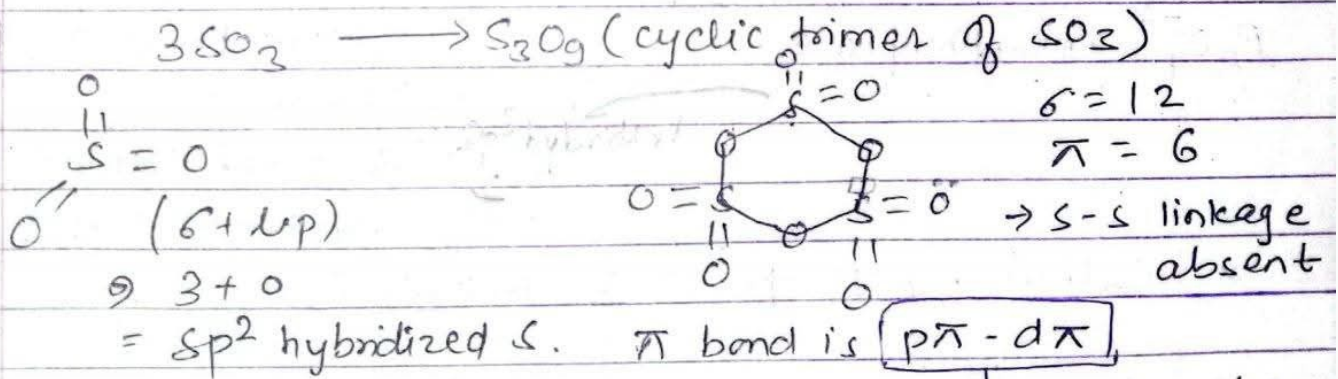
- Effect of Heat on Acid:



fumes of SO_2

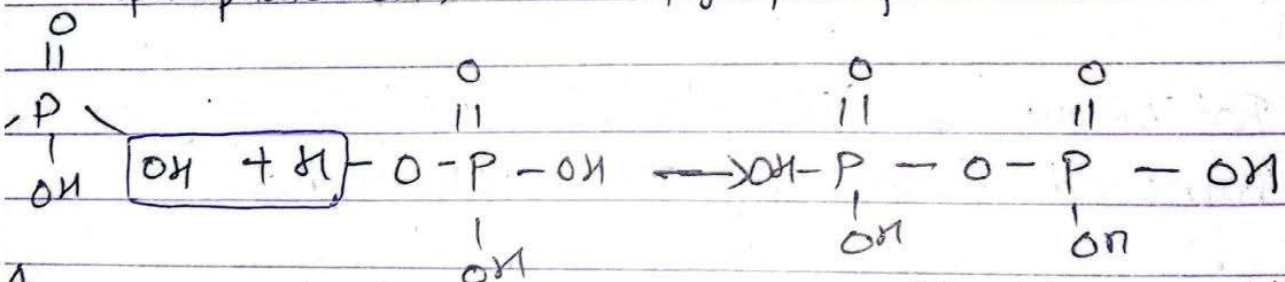
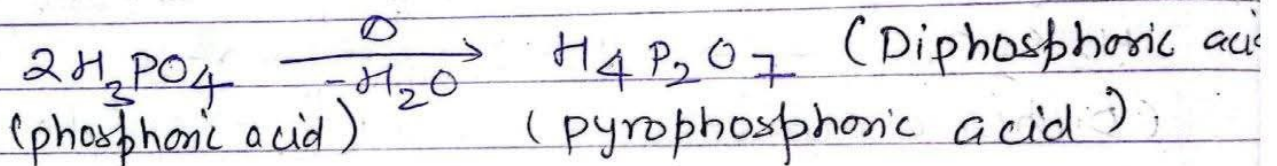
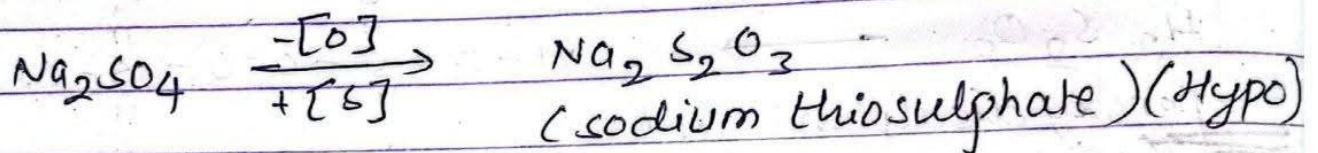
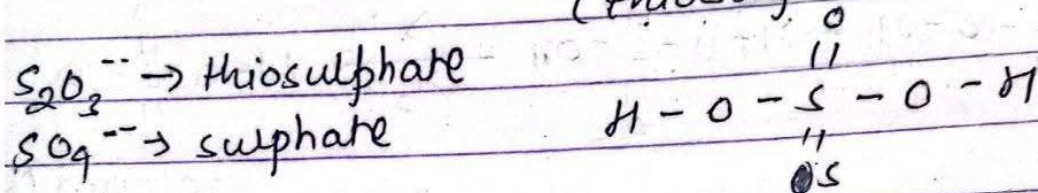
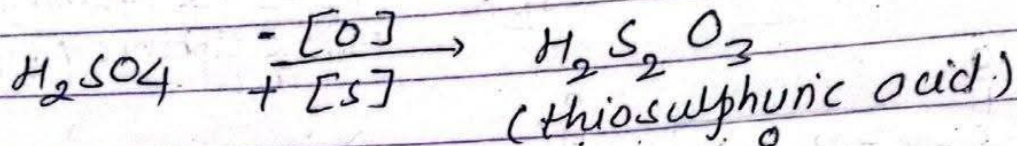
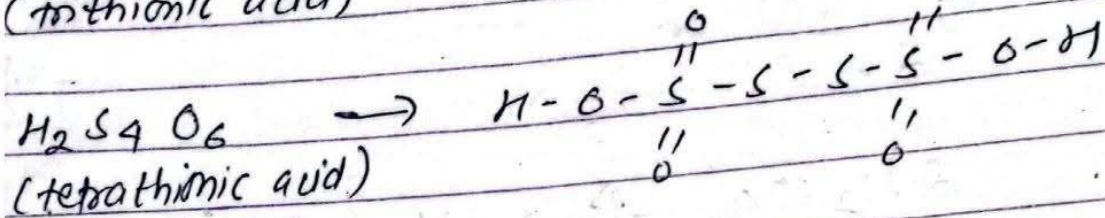
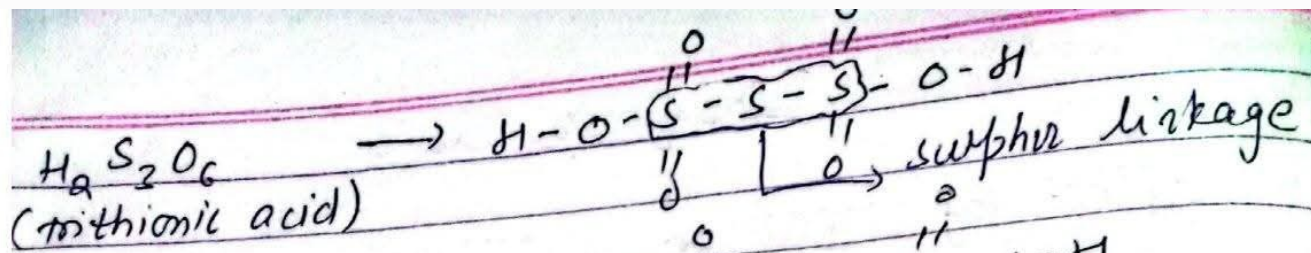


For below O_6 like $\text{H}_2\text{S}_2\text{O}_5$, $\text{H}_2\text{S}_2\text{O}_4$, oxygen should be removed from any π -bonded sulphur.



↓

bond formed by the vacant d-orbital of sulphur and lone pair of electron in p-orbital of oxygen.



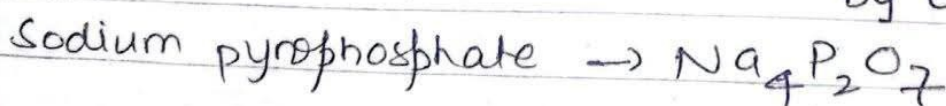
4 $\rightarrow$ pyrophosphate

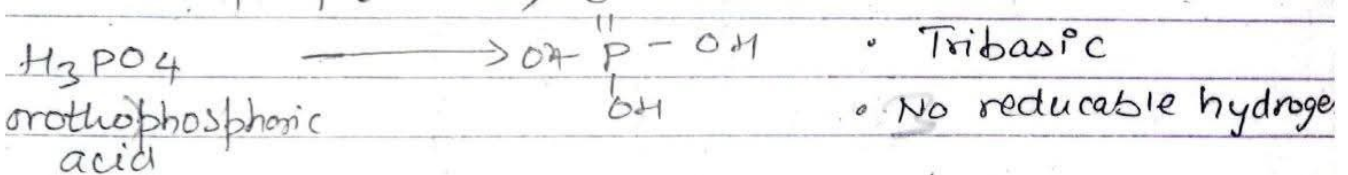
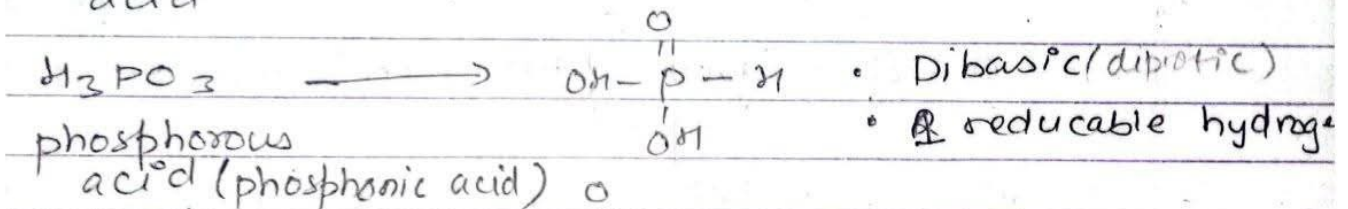
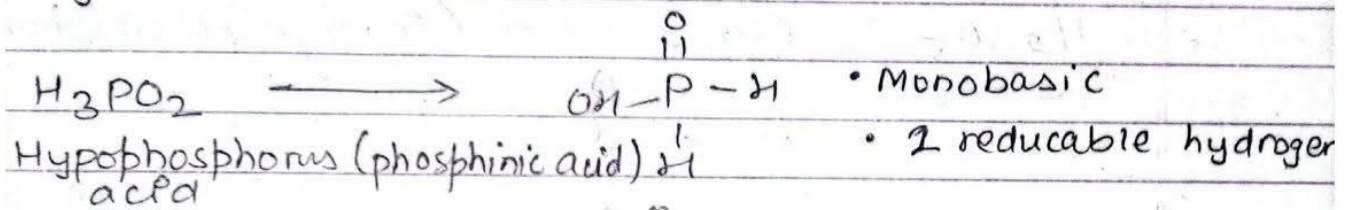
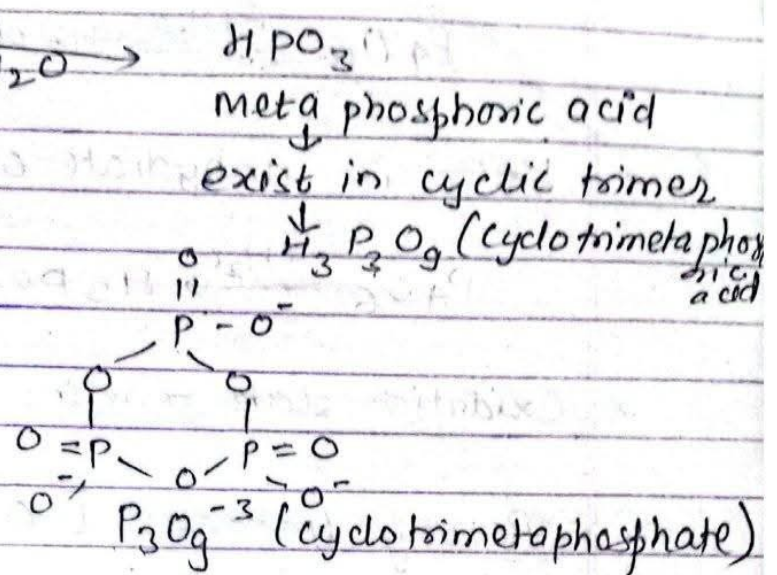
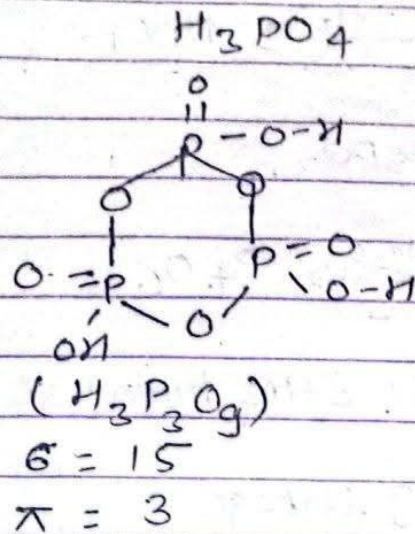
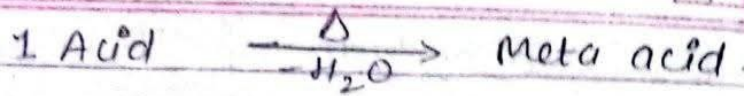
can release
4 hydrogen

Pyrophosphoric acid

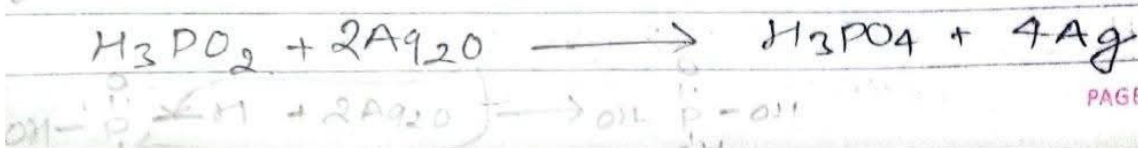
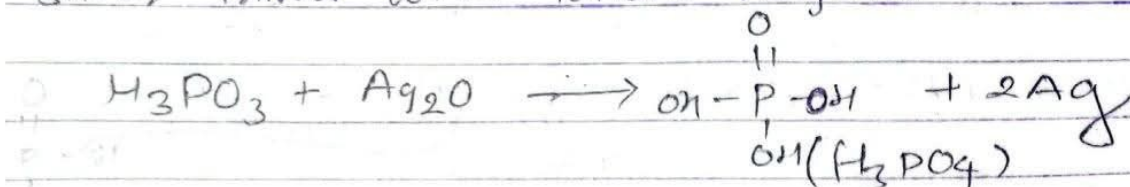
$\rightarrow$ Tetrabasic

$\rightarrow$ 2-tetrahedral unit joined
by common oxygen

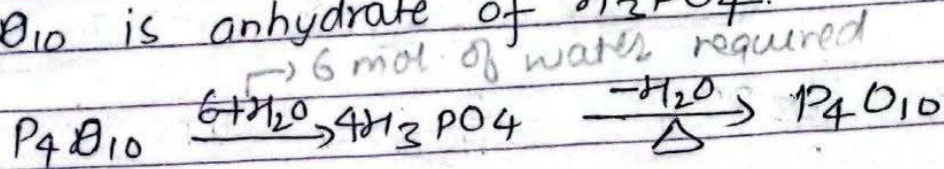




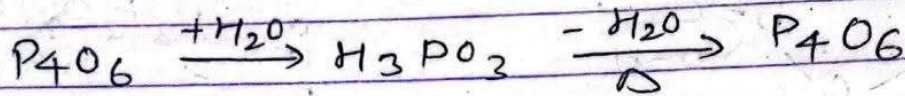
H_3PO_2 and H_3PO_3 acts as a reducing agent due to presence of reducible hydrogen (P-H), and gives silver mirror with Tollen's reagent.



P_4O_{10} is anhydride of H_2PO_4 .



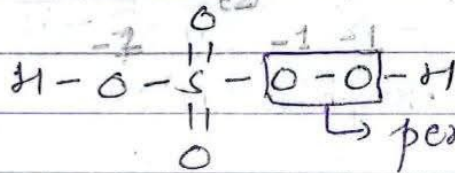
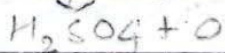
P_4O_6 is anhydride of H_3PO_3 .



Oxidation state remains same after hydrolysis (i.e. +5)

Peroxy linkage: $[O^{\ominus} - O^{\ominus}]$ linkage

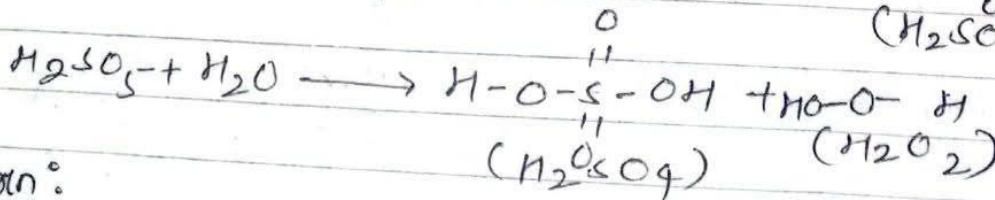
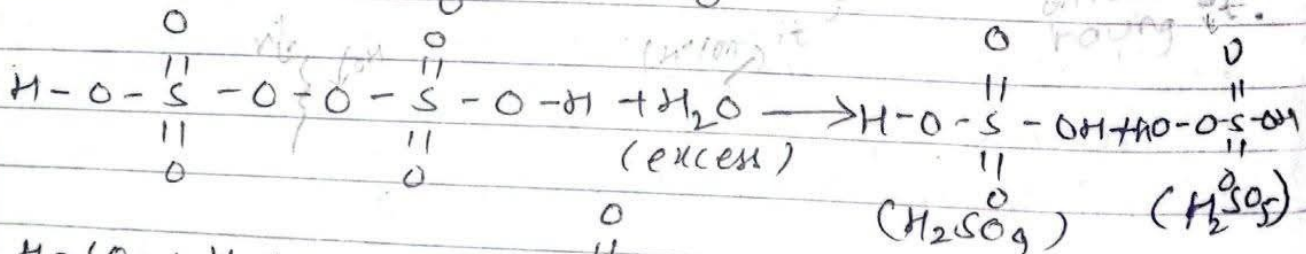
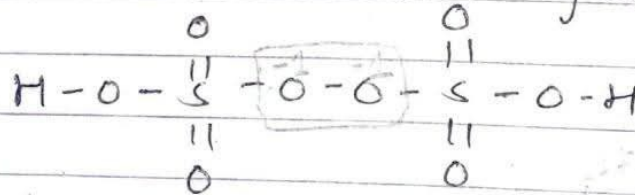
H_2SO_5 : Caro's acid (Peroxymonosulphuric acid)



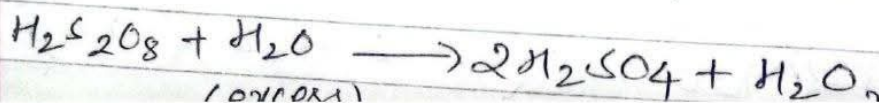
→ peroxide linkage

O.S. of sulphur = +6, sp^3 -hybridized

$H_2S_2O_8$: Marshall's acid (Peroxodisulphuric acid)



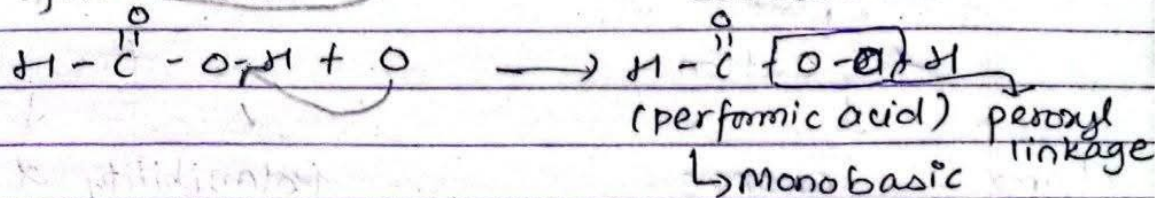
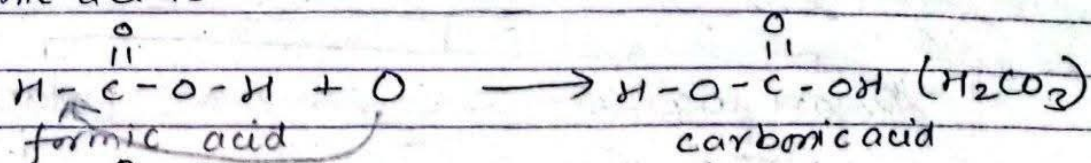
an:



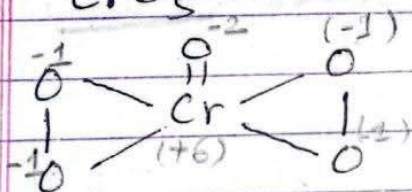
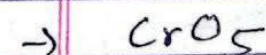
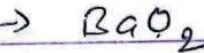
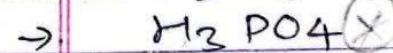
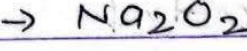
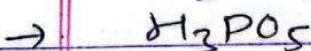
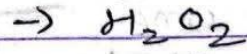
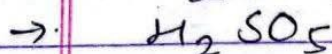
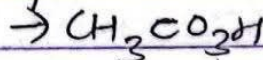
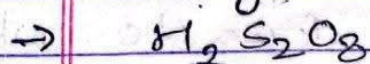
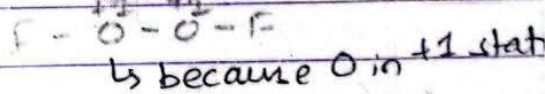
oxide

 O_2
peroxide O_2
superoxide $S > P > O$ (oxidizing power)DATE

• Performic acid:

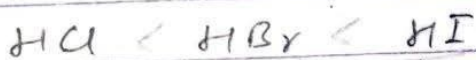
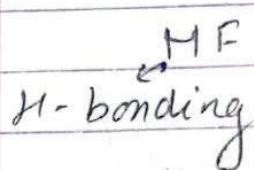


• Peroxy linkage is present in:

• Peroxy linkage absent in O_2F_2 

• which is highly volatile?

a) HI b) HF c) HBr d) HCl

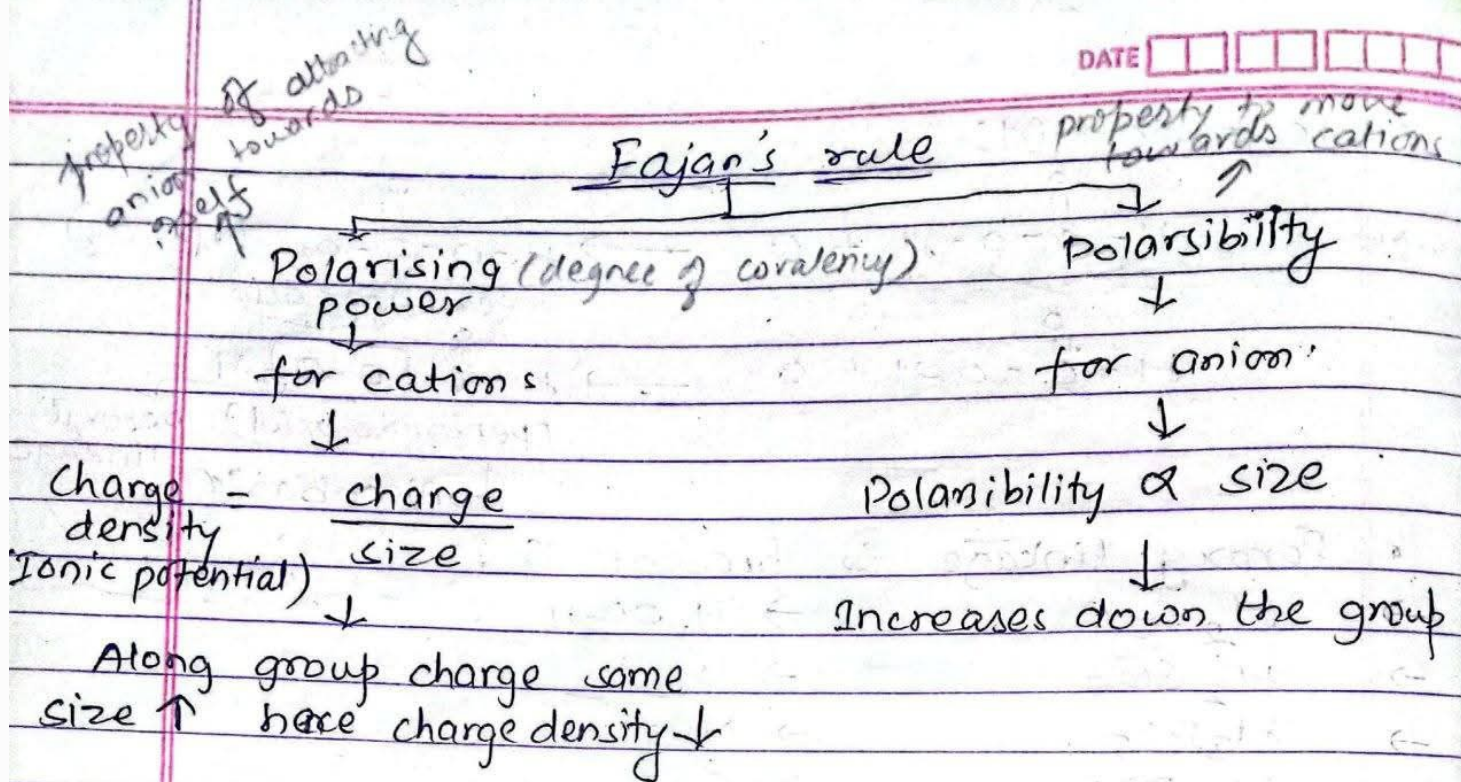


$$\text{B.P} \propto \frac{1}{\text{volatility}}$$

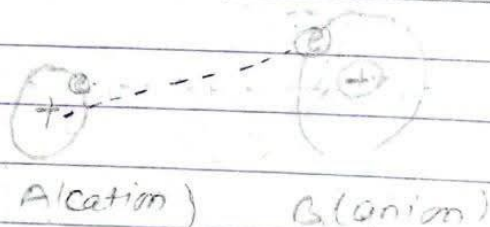
$$\text{vanderwal Force} \propto m$$
order of Boiling point: $\text{HF} > \text{HI} > \text{HBr} > \text{HCl}$ volatility $\propto$: $\text{HCl} > \text{HBr} > \text{HI} > \text{HF}$ Acidic strength: $\text{HI} > \text{HBr} > \text{HCl} > \text{HF}$ * BP: $\text{BiH}_3 > \text{SbH}_3 > \text{NH}_3 > \text{AsH}_3 > \text{PH}_3$

weak H-bond

PAGE



- Experimentally 100% covalent or 100% ionic bond is not possible.
- Covalent character in ionic compound is explained by polarisation which was explained by Fajan.

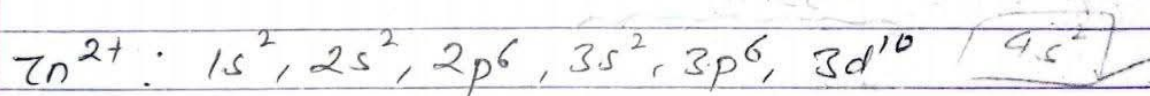
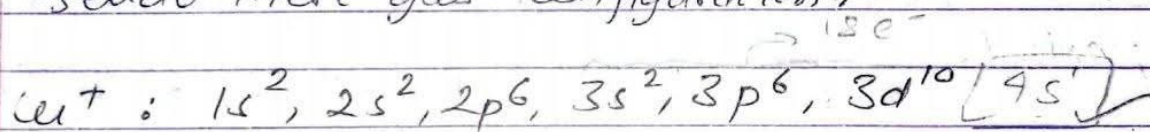


- According to Fajan rule, when ionic bond is formed, cation and anion come close to each other in isolated condition.
- Then e^- cloud of anion is attracted by charge of cation. Similarly e^- cloud of cation is attracted by nucleus of anion. As a result distortion is created in e^- cloud of both.

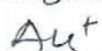
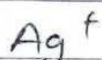
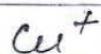
ions but distortion of e^- cloud of cation is negligible because of its small size (ie e^- cloud of cation is strongly hold by its nucleus).

- Distortion in e^- cloud of anion by cation is called polarisation. The tendency of cation to distort anion is called polarising power of cation.
- The ability of anion to get polarised called polarisibility.
- charge of cation $\uparrow$, polarising power $\uparrow$, covalent character $\uparrow$
- Size of cation $\uparrow$, PP $\downarrow$, covalent character $\downarrow$
- Size of anion $\uparrow$, polarisibility $\uparrow$, covalent character $\uparrow$
- Charge of anion $\uparrow$, Polarisibility $\uparrow$, covalent character $\uparrow$

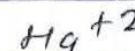
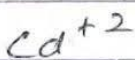
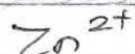
Pseudo inert gas configuration:



IB (11^{th} group)



IIIB (12^{th} group)



pseudo inert gas configuration

Polarising power increases down the group for pseudo inert gases, just opposite of others.

- Polarising power is maximum for pseudoinert gases.



Seen if charge & shape/size are almost equal.

Eg: Na^+Cl^- & Cu^+Cl^- , K^+I^- & Ag^+I^-
 same charge & size

Application of Fajan Rule:

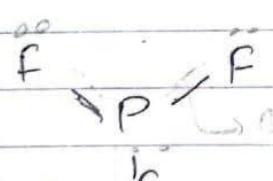
- Polarisation $\uparrow$, Covalent character $\uparrow$, I.C. $\downarrow$, M.P. $\downarrow$ and solubility in H_2O $\downarrow$.

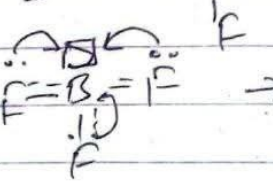
i.e. Covalent character $\propto \frac{1}{\text{Ionic character}} \propto \frac{1}{\text{M.P.}} \propto \frac{1}{\text{solubility in } H_2O}$

Back Bonding:

Conditions:

- One bonded atom must have lone pair and other bonded atom must have vacant orbital.
- One bonded atom must be of 2nd period & other bonded atom must be of 2nd or 3rd period.

Exeg. PF_3  $P = 1s^2, 2s^2, 2p^6, 3s^2, 3p^3, 3d^0$
 $\rightarrow$ back bonding
 $\rightarrow$ vacant orbital

• BF_3  $\rightarrow$ vacant p orbital of boron and lone pair of fluorine

Because of back bonding there is development of partial double bond.

SiH_3 $\rightarrow$ vacant 3d present in Si

$H_3Si - \overset{\cdot\cdot}{\underset{\cdot\cdot}{N}} = SiH_3$
 $\rightarrow$ lone pair

Trisilyl amine

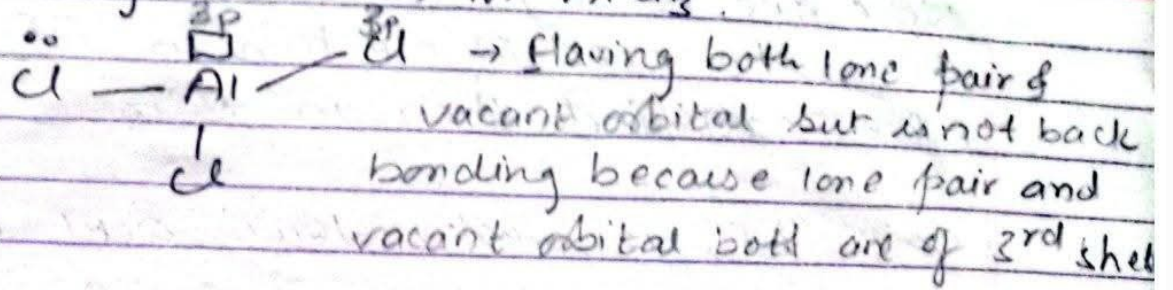
Hybridization of N

$$= lp + 6$$

$$= 3 + 0 = sp^2$$

(lone pair $\downarrow$ in back bonding not counting)
 PAGE

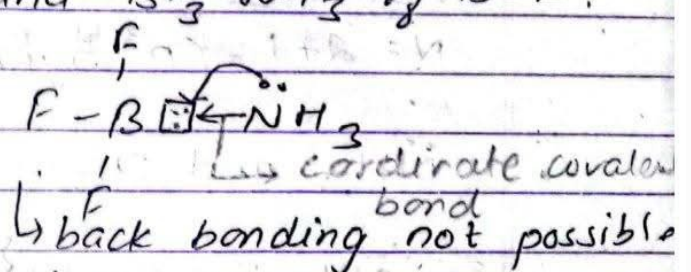
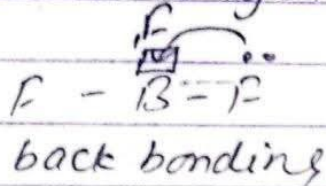
- Backbonding not seen in $AlCl_3$.



* Which molecule is planar?

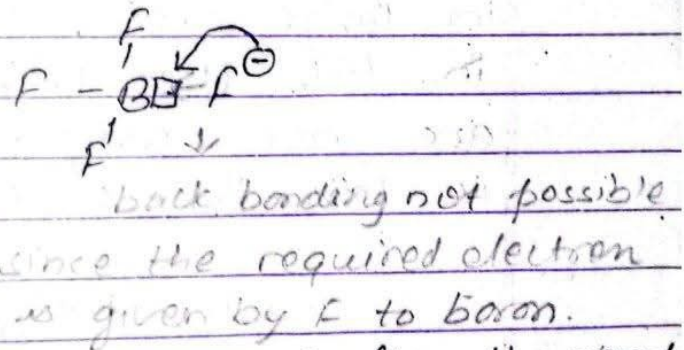
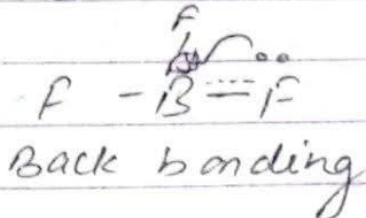
- a) $(CH_3)_2N$ b) $(SiH_3)P$ ☒ c) $(SiH_3)_3N$ d) b and c

- Bond length of BF_3 and $BF_3 \cdot NH_3$ of B-F.



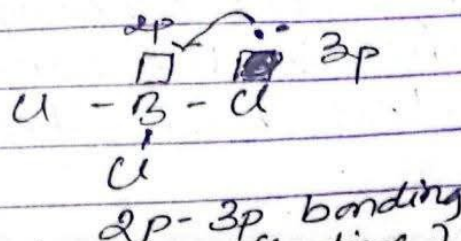
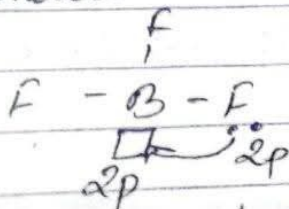
$\therefore$ Bond length: $BF_3 < BF_3 \cdot NH_3$ since vacant orbital is occupied

- BF_3 and BF_4^-



- Backbonding completely does not satisfy the need of electron whereas coordinate bond does.

- Effective back-bonding is seen in similar size orbitals.

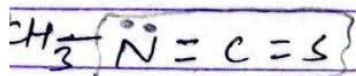


Lewis acid $\rightarrow$ Accepts pair of e⁻

DATE

Since the bonding is more effective in BF_3 than BCl_3 , the need of electron in BF_3 will be more satisfied than in BCl_3 .

So, BF_3 is weaker Lewis acid than BCl_3 which need more electron

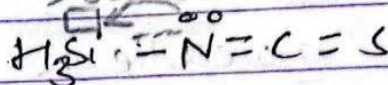


isocyanate

methyl isocyanate

no back bonding

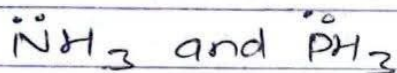
$\text{N} = 2 + 1 = \text{sp}^3$ hybridized



silyl isocyanate

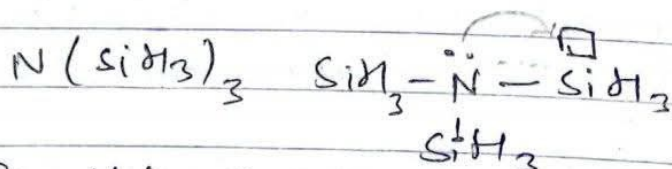
back bonding possible

$\text{N} = 0 + 2 = \text{sp}^2$



In $\ddot{\text{N}}\text{H}_3$ the electrons are dispersed in very small area but in $\ddot{\text{P}}\text{H}_3$, the electrons are dispersed in large area, so the electrons are stable in $\ddot{\text{P}}\text{H}_3$ than $\ddot{\text{N}}\text{H}_3$. So, the lone pair electrons are not ready to bind with H^+ are less basic than $\ddot{\text{N}}\text{H}_3$.

Basic strength: $\ddot{\text{P}}\text{H}_3 < \ddot{\text{N}}\text{H}_3$



In this compound the lone pair electrons undergo back-bonding hence are not available for H^+ . So the compounds going back-bonding are less basic than others.

- Acidic Lewis acidic order of AlCl_3 , AlBr_3 & AlI_3 :

Since none the above compound undergo back-bonding so, in this case we observe electronegativity of central or side atom.

LA $\propto$ EN of central / side atom.

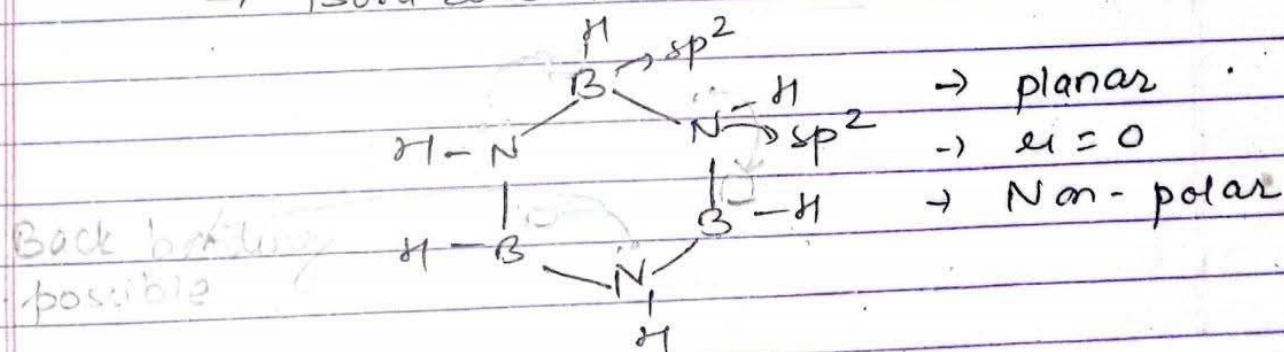
Here central atom Al is common in all but EN is of order $\text{Cl} > \text{Br} > \text{I}$.

So, L.A order: $\text{AlCl}_3 > \text{AlBr}_3 > \text{AlI}_3$

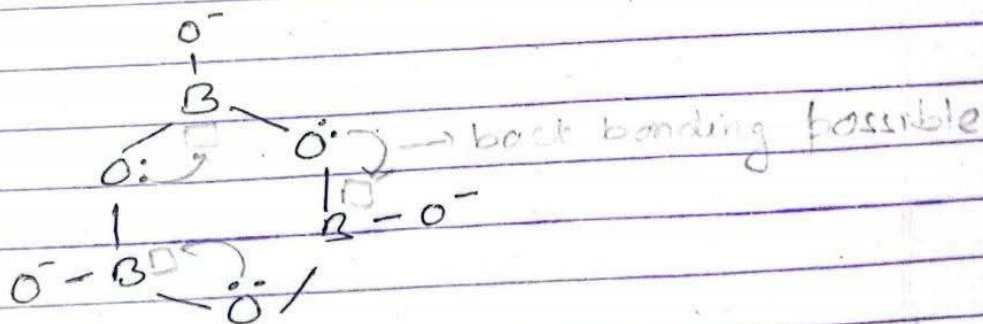
- $\text{BF}_3 < \text{BCl}_3 < \text{BBr}_3 < \text{BI}_3$ [Lewis Acid]

$\text{B}_3\text{N}_3\text{H}_6$: Inorganic benzene

$\rightarrow$ Borazole, Borazine



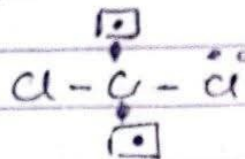
$\text{B}_3\text{O}_6^{-3}$:



- CCl_2 exist in two forms $\left\{ \begin{array}{l} \rightarrow \text{singlet} \\ \rightarrow \text{triplet} \end{array} \right.$



singlet
backbonding
possible



triplet $\rightarrow$ backbonding
not possible

Hence, $a > b$ [stability]

- CCl_2 and CCl_3



Backbonding from $\text{Cl} \rightarrow \text{C}$.
($3p\pi - 2p\pi$)



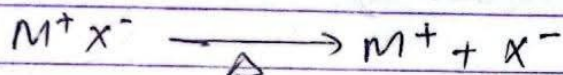
Back bonding from $\text{C} \rightarrow \text{Cl}$.
($2p\pi - 3d\pi$)

Thermal stability

Polyatomic anion: CO_3^{2-} , SO_4^{2-} , NO_3^- , HCO_3^- , OH^- , O_2^{2-} , O_2^- .

Monoatomic anion: X^- , O^{2-} , N^{3-} , H^-

- Thermal stability of ionic compound having monoatomic anion

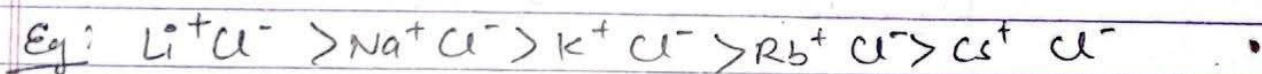


Thermal energy decided by Lattice energy.

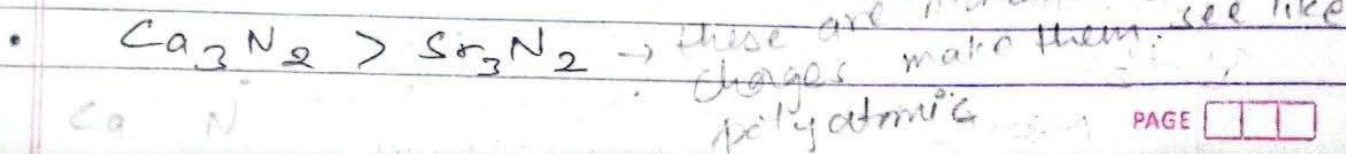
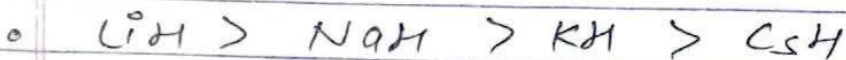
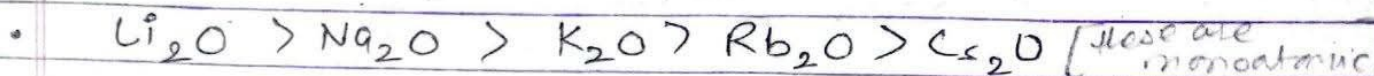
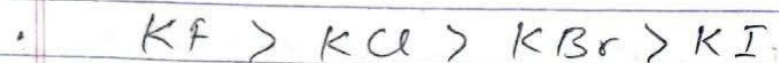
$$\text{Lattice energy} = \frac{K \cdot q_1 q_2}{r(r_c + r_a)}$$

$$\text{L.E} \propto \text{charge} \quad \& \quad \text{LE} \propto \frac{1}{r(r_c + r_a)}$$

Thermal stability $\propto$ Lattice energy

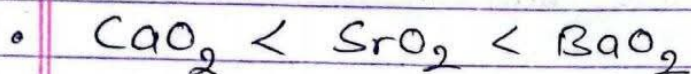
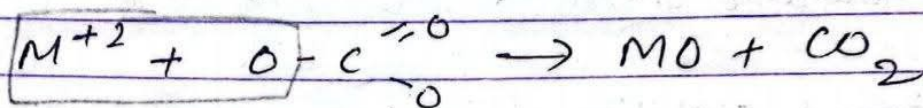
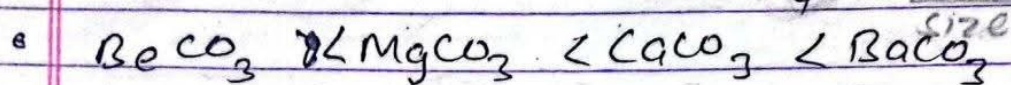


Same charge, so then check size, smaller size greater lattice energy and greater is thermal stability.



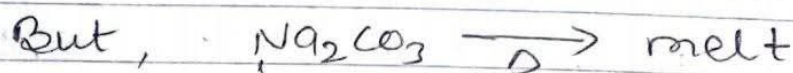
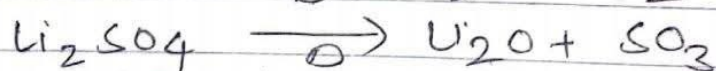
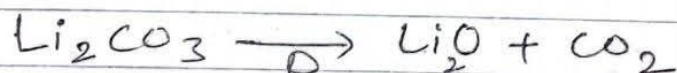
- Thermal stability of polyatomic anion :

increases
down the group Thermal stability $\propto \frac{1}{\phi}$ [ϕ = Polarising power]
 $\phi = \frac{\text{charge}}{\text{size}}$



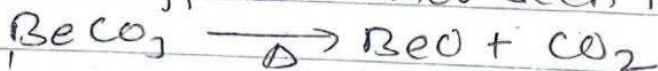
Thus, thermal stability increases down the group.

* carbonates and sulphates of Group IA do not break to give respective compounds rather melt except lithium.



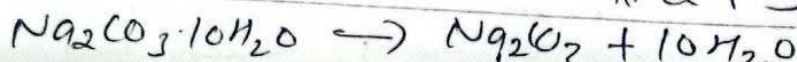
so on for
sulphates

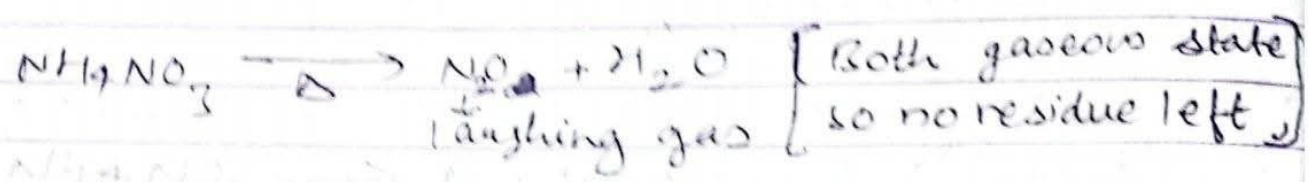
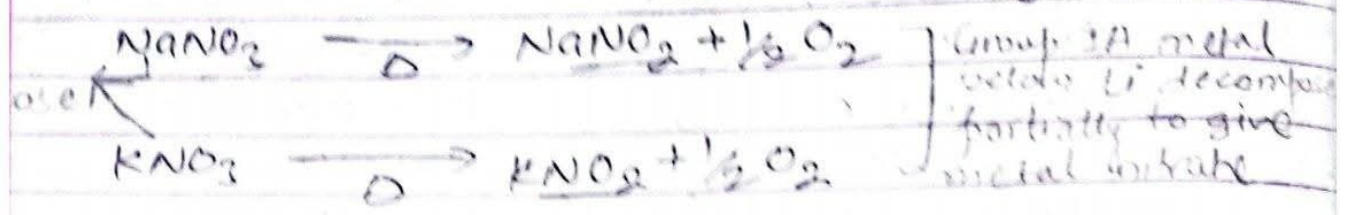
* But such effect is not seen in Group IIA.



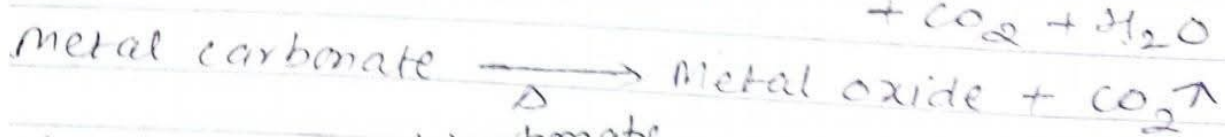
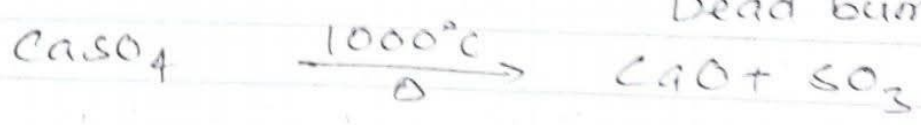
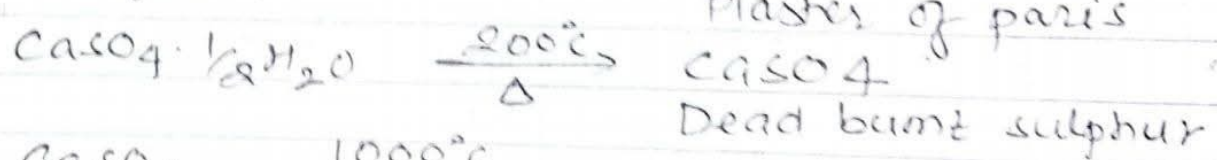
- washing soda loses wt on heating due to

a) H_2O b) CO_2 c) CO d) a & b





Effect of heat on Gypsum:



Group IA metals ^{bicarbonate} except Li exist as solid state

$\text{NaHCO}_3, \text{KHCO}_3, \text{RbHCO}_3, \text{CsHCO}_3$ - solid state
 $\text{LiHCO}_3 \rightarrow$ solution state

Group IIA metals exist in solution state

Stability

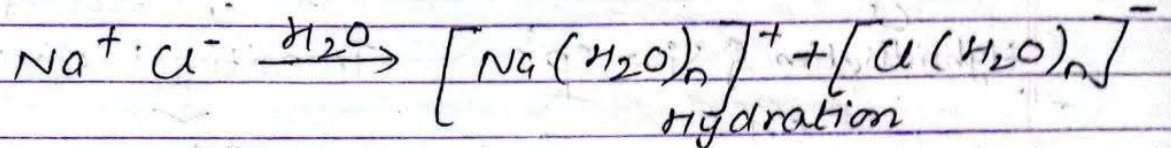
- $H_2O > H_2S > H_2Se > H_2Te$
 - $HF > HCl > HBr > HI$
 - $NH_3 > PH_3 > AsH_3 > SbH_3$
- Because of bond length.
shorter bond length, greater stability

DATE

Hydrolysis :

surrounding the atom by H_2O molecule.

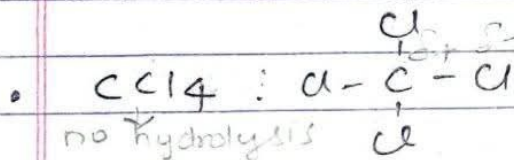
- Hydrolysis of halide $\propto$ covalent character
- Ionic chloride undergo hydration process.



But covalent chloride hydrolysed by H_2O .

Condition for undergoing hydrolysis:

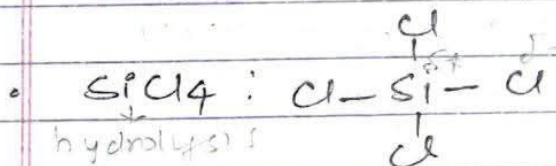
- Vacant orbital and +ve charge present simultaneously at same atom.



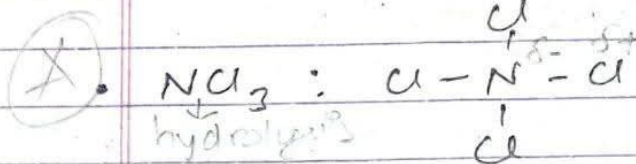
$C = 1s^2, 2s^2, 2p^2$ $\uparrow\downarrow$ $\uparrow\downarrow$ $\uparrow\downarrow$

1s 2s 2p

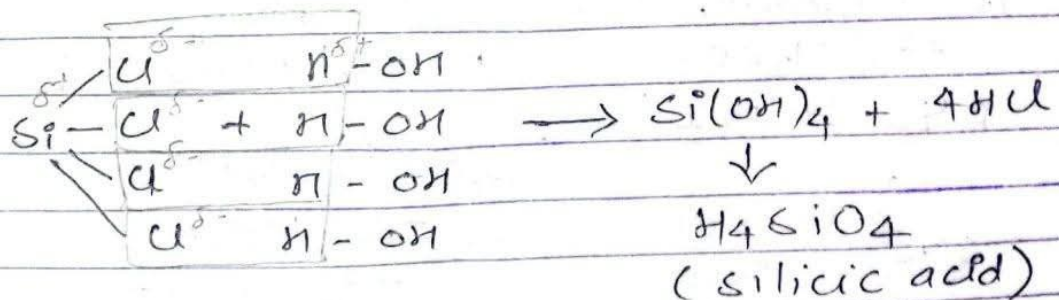
no vacant orbital present
hence no hydrolysis.



undergo hydrolysis because of present of 3d-vacant orbit



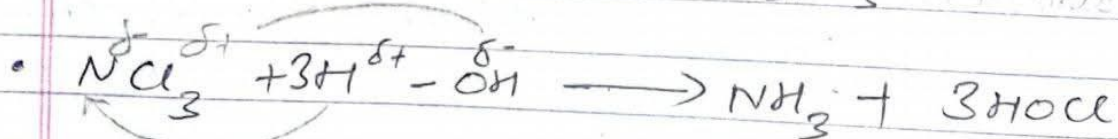
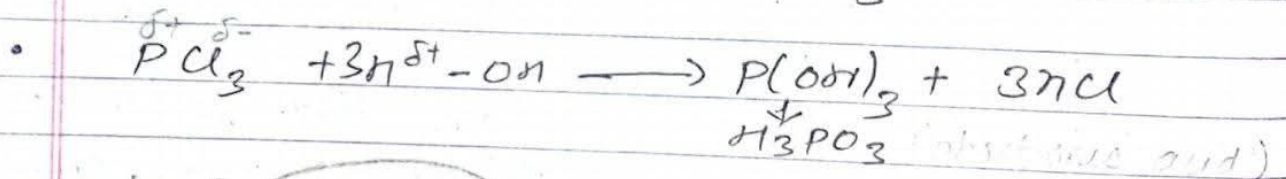
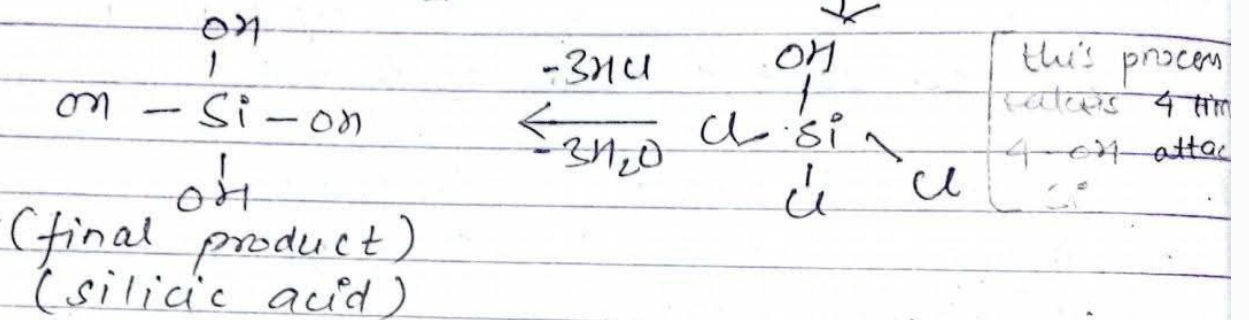
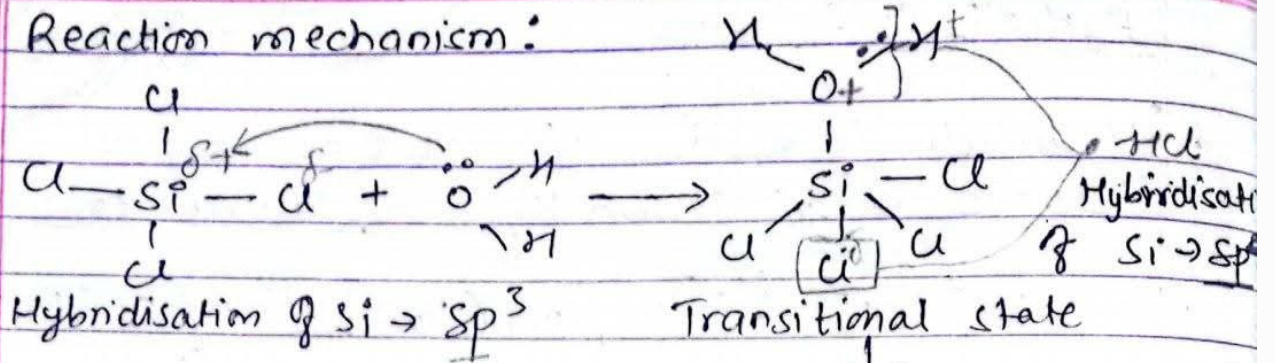
EN of N $\approx$ EN of Cl, but because of small size of N it has a little more EN than N so get -d.



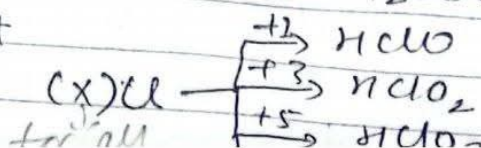
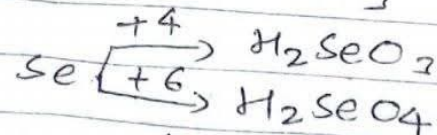
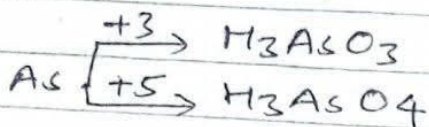
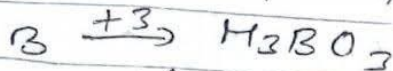
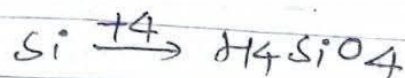
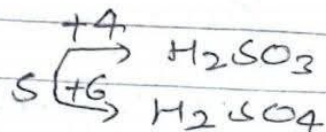
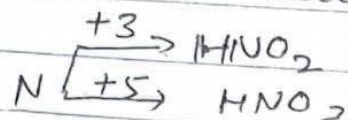
Note: If acid has oxygen and hydrogen, it is called oxyacid.

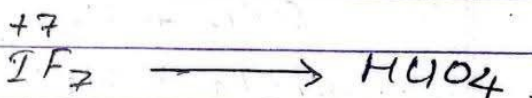
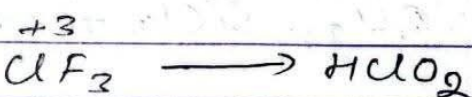
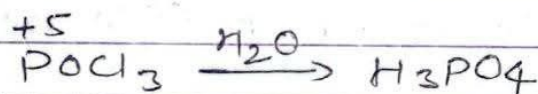
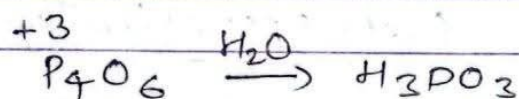
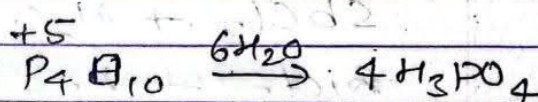
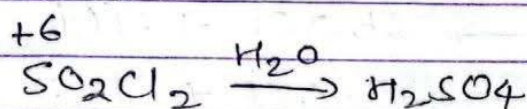
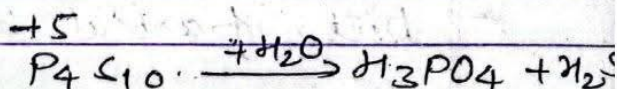
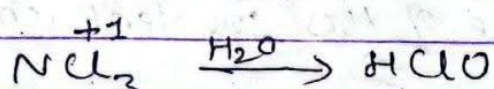
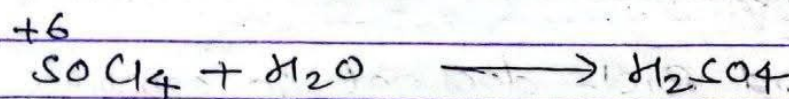
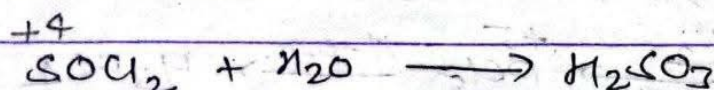
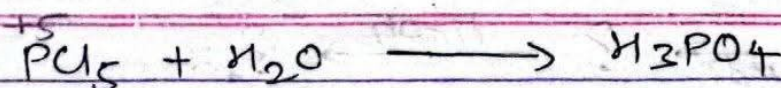
Hybridization in intermediate product is always +1 than the reactant. Eg: reactant sp^3 , intermediate sp^3d

Reaction mechanism:

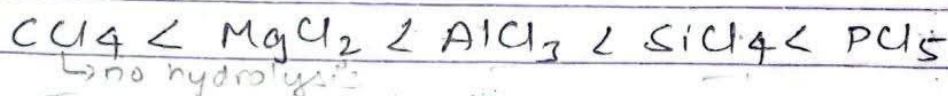
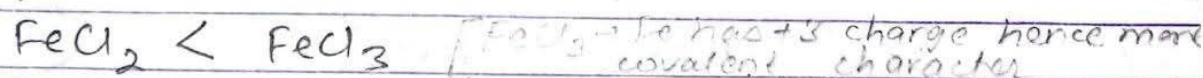
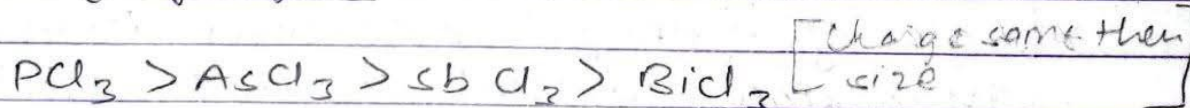


Key points of hydrolysis: The result of hydrolysis is oxyacids.

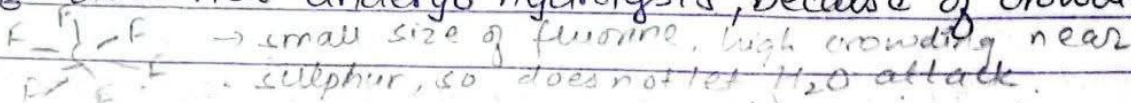




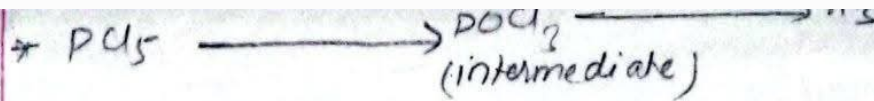
Rate of hydrolysis $\propto$ covalent character.



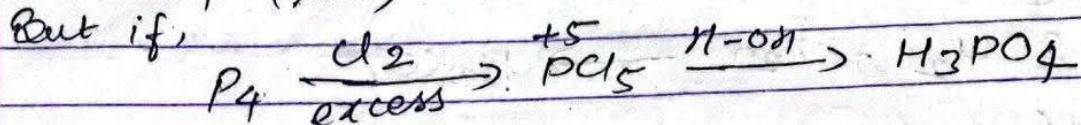
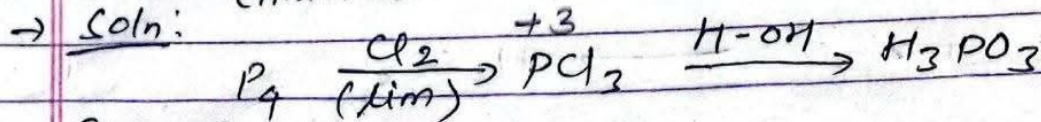
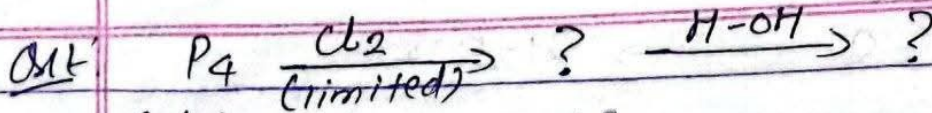
SF_6 does not undergo hydrolysis; because of crowding



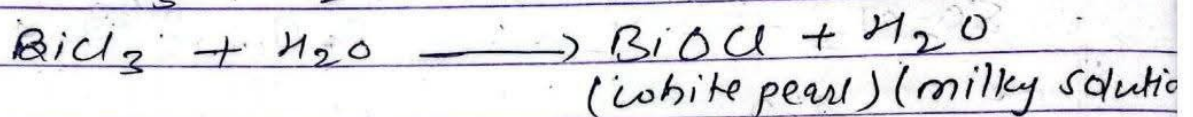
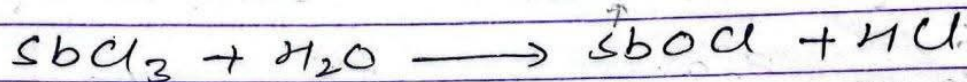
SCl_6 , $\text{SBr}_6 \rightarrow$ undergo hydrolysis because of larger size



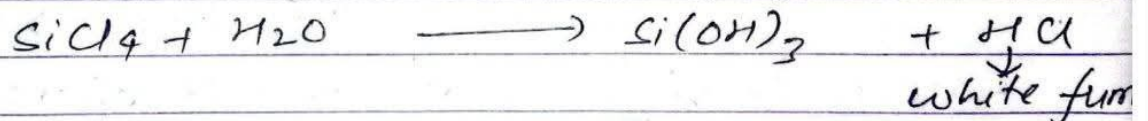
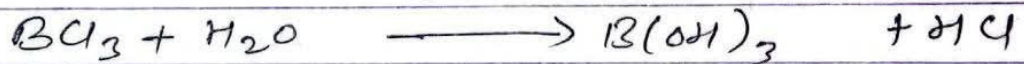
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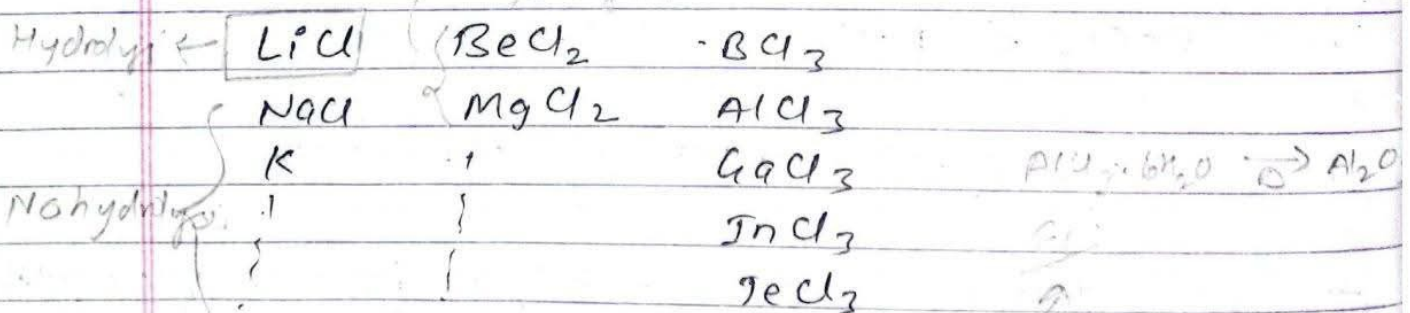
* $SbCl_3$ and $BiCl_3$ does not go complete hydrolysis but partial because of less covalent character.



* BCl_3 , $SiCl_4$ fumes in air but CCl_4 not.
means moisture

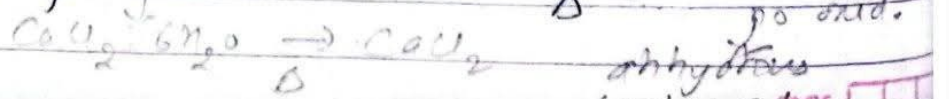


* Rate of hydrolysis decreases down the group since covalent character decreases.

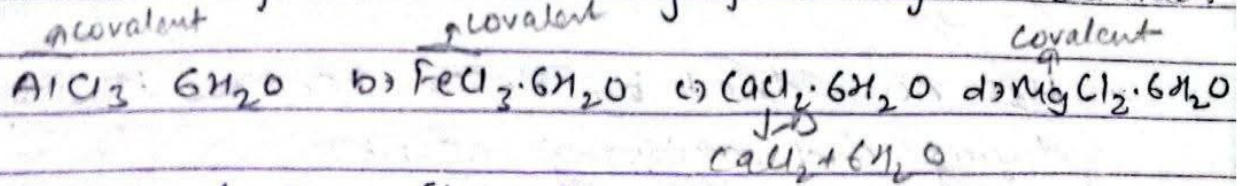


• If covalent compound has water $\xrightarrow{\Delta}$ metal oxide

Ionic compound has water $\xrightarrow{\Delta}$ water lost

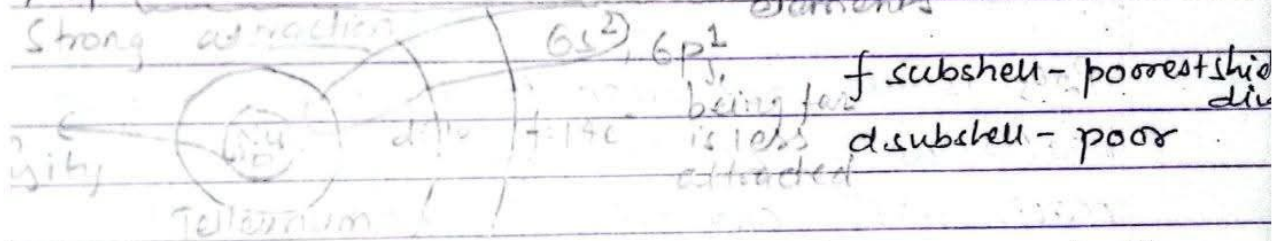


which compound on heating form anhydrous chloride?



Inert pair effect : p-block element

2p ¹ B	C	N	Al → 3s ² , 3p ¹ ↓ it can lose only one electron to get +1 o.s. or lose all 3 electron to get +3 state. only applicable to these elements
3p ¹ Al	Si	P	
4p ¹ Ga	Ge	As	
5p ¹ In	Sn	Sb	
6p ¹ Tl	Pb	Bi	



Due to poor shielding effect of f + d subshell, the electrons beyonds are easily / strongly attracted towards nucleus. As above eg: 6s² electron is tightly hold by nucleus and is not available for bonding, so only electron of 6p¹ takes part in bonding.

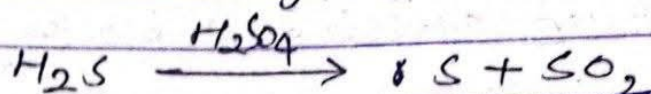
Eg: PbCl_2 is more stable than PbCl_4 , But SnCl_4 is more stable than SnCl_2 because it does not show inert pair effect.

$\text{Cl}_2 \rightarrow$ dried by conc H_2SO_4

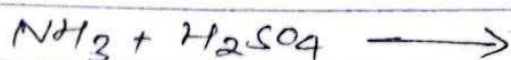
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Drying agent:

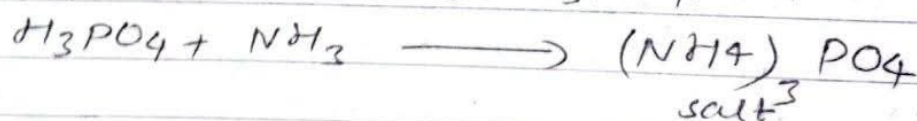
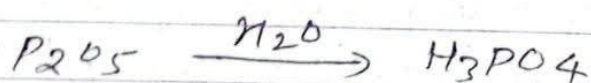
H_2S not dried by H_2SO_4 because H_2SO_4 is drying and oxidising agent.



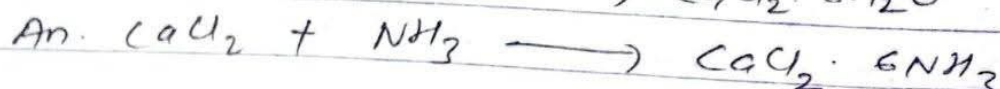
NH_3 is not dried by H_2SO_4 , because H_2SO_4 being acid reacts with Base NH_3 to form salt.



P_2O_5 is also not used to dry NH_3 , because, P_2O_5 reacts with moisture forming H_3PO_4 , which reacts with ammonia to form salt.



Anhydrous CaCl_2 also cannot be used because it along with absorbing H_2O reacts with NH_3 .



So NH_3 is dried by CaO , because it reacts with H_2O to form Ca(OH)_2 which being base does not react with NH_3 .

